Q1.

The kinetic energy, E, of a compound pendulum is given by

$$E = \frac{1}{2}I\omega^2$$

where ω is the angular speed and I is a quantity called the moment of inertia.

(a) Show that for this formula to be dimensionally consistent then I must have dimensions ML^2, where M represents mass and L represents length.

(2)

(b) The time, T, taken for one complete swing of a pendulum is thought to depend on its moment of inertia, I , its weight, W, and the distance, h, of the centre of mass of the pendulum from the point of suspension.

The formula being proposed is

$$T = kI^{\alpha} W^{\beta} h^{\gamma}$$

where k is a dimensionless constant.

Determine the values of α, β and γ.

(3)

(Total 5 marks)

Q2.

A tank containing a liquid has a small hole in the bottom through which the liquid escapes. The speed, u m s^{-1}, at which the liquid escapes is given by

$$u = C\,V\rho\,g$$

where V m^3 is the volume of the liquid in the tank, ρ kg m^{-3} is the density of the liquid, g is the acceleration due to gravity and C is a constant.

By using dimensional analysis, find the dimensions of C.

(Total 5 marks)

Q3.

The magnitude of the gravitational force, F, between two planets of masses m_1 and m_2 with centres at a distance x apart is given by

$$F = \frac{Gm_1m_2}{x^2}$$

where G is a constant.

(a) By using dimensional analysis, find the dimensions of G.

(3)

(b) The lifetime, t, of a planet is thought to depend on its mass, m, its initial radius, R, the constant G and a dimensionless constant, k, so that

$$t = km^{\alpha} R^{\beta} G^{\gamma}$$

where α, β and γ are constants.

Find the values of α, β and γ.

(5)
(Total 8 marks)

Q4.

The time T taken for a simple pendulum to make a single small oscillation is thought to depend only on its length l, its mass m and the acceleration due to gravity g.

By using dimensional analysis:

(a) show that T does **not** depend on m;

(3)

(b) express T in terms of l, g and k, where k is a dimensionless constant.

(4)
(Total 7 marks)

Q5.

An expression for calculating the volume per second of liquid flowing through a cylindrical pipe is of the form

$$kl^{-1}r^{a}p^{b}\eta^{c}$$

where k is a dimensionless constant;
l metres is the length of the pipe;
r metres is the radius of the pipe;
p is the excess pressure, measured in N m^{-2};
and η is the viscosity of the liquid measured in kg m^{-1}s^{-1}.

(a) (i) Write down the dimensions of η.

(1)

(ii) Determine the dimensions of p.

(2)

(b) By using dimensional analysis, find the values of a, b and c.

(6)
(Total 9 marks)

Q6.

The gravitational force exerted between two spheres of masses m_1 and m_2 is

$$\frac{Gm_1m_2}{r^2}$$

where G is a constant and r is the distance between the centres of the two spheres.

(a) Find the dimensions of G.

(3)

(b) The speed, v, of a satellite orbiting a planet of mass m is given by

$$v = \frac{G^{\alpha} m^{\beta}}{r^{\gamma}}$$

where r is the radius of the orbit and α, β and γ are constants. Find the values of α, β and γ for this equation to be dimensionally consistent.

(4)

(Total 7 marks)

Q7.

The acceleration, a, of a body falling with speed v and subject to air resistance may be modelled by the equation

$$a = g - \lambda v^2$$

where λ is constant.

Find the dimensions of λ in order that the equation is dimensionally consistent.

(Total 4 marks)

Q8.

The tension, T, in a stretched elastic string is given by Hooke's Law, $T = \frac{\lambda x}{l}$, where λ is the modulus of elasticity, l is the natural length of the string and x is the extension of the string.

Find the dimensions of λ.

(Total 4 marks)

Q9.

The gravitational force of the sun, which has mass m_1, on a planet, of mass m_2, is an attractive force directed along the line joining them and of magnitude $\frac{Gm_1m_2}{d^2}$ where d is the distance between their centres and G is the universal gravitational constant.

Use dimensional analysis to find the dimensions of G.

(Total 4 marks)